

IKRAM Ullah

ALL PROJECTS — DETAILED PORTFOLIO

Academic • AI/ML • Deep Learning • NLP • Software • Robotics • Data Engineering • SDR

A consolidated record of projects and technical work based on the project files and personal project records available in the user's workspace.

Document rule: documented work is expanded; projects with limited evidence are explicitly marked instead of being filled with invented details.

Project Index

- 1. AI-Powered Resume Screening System
- 2. Real-Time Data Warehouse Architectures for Predictive Analytics in Formula 1
- 3. Human Activity Recognition Using Deep Learning
- 4. KRR for Robotics — KnowRob, RoboBrain & CRAM
- 5. Amazon Reviews NLP Assignment
- 6. Ikrambot — Personal RAG Chatbot
- 7. Online Event Management & Ticket Booking System
- 8. eCommerce Inventory Management System / DSA Projects
- 9. Robotics & Embedded Systems Projects
- 10. RTL-SDR & Weather Satellite Exploration
- 11. Professional Communication Portfolio
- 12. Hospital LAN Design
- 13. MyFitGoal Flutter Fitness App
- 14. Human Pose Estimation
- 15. GAN Research

1. AI-Powered Resume Screening System

Machine Learning Semester Project

Area	Details
Status	Academic ML project / documented implementation
Scope note	The available records support the architecture, tools and workflow below. Some operational details were design requirements rather than validated production results.

Objective

Automate parts of HR resume screening by extracting resume text, predicting resume categories, comparing resumes with a job description, and supporting screening/notification workflows.

Workflow

1. Resume input from PDF/DOCX.
2. Text extraction and preprocessing.
3. TF-IDF feature extraction with word n-grams.
4. LinearSVC classification.
5. LabelEncoder for target labels.
6. Cosine similarity between resume text and job description.
7. Streamlit user interface.
8. Gmail-based resume handling and notification workflow.

Technical stack

Area	Details
Feature extraction	TF-IDF; max_features=5000; ngram_range=(1,2)
Classifier	LinearSVC
Labels	LabelEncoder
Matching	Cosine similarity
Interface	Streamlit
Documents	PDF/DOCX
Email	Gmail integration

Project files / artifacts

- tfidf.pkl
- model.pkl
- encoder.pkl
- Resumes/
- job_description.txt
- ResumeDataSet.csv

Dataset context

A documented dataset version named ResumeDataSet.csv contained approximately 962 rows and two main columns: Category and Resume. Other dataset versions contained fields such as ID, Resume_str, Resume_html and Category.

Intended screening flow

9. HR supplies a job description.

10. Resumes are collected.
11. Text is extracted.
12. The classifier predicts a category.
13. Cosine similarity provides a JD-matching signal.
14. The results support screening/shortlisting.
15. A previously discussed 75% threshold was a project rule for shortlist/notification workflow, not a validated hiring standard.

Issues encountered

- A Gmail test was recorded where zero resumes were fetched even though an unread DOCX had been sent.

Skills

Python, ML preprocessing, TF-IDF, LinearSVC, label encoding, cosine similarity, document processing, Streamlit and Gmail integration.

2. Real-Time Data Warehouse Architectures for Predictive Analytics in Formula 1

Data Warehouse / Streaming / Predictive Analytics

Area	Details
Status	Academic project / research-heavy architecture
Scope note	The project combines FastF1 data, streaming architecture and predictive models. Advanced live-system elements should be described as architecture/research unless separately verified as deployed.

Title and case

Real-Time Data Warehouse Architectures for Predictive Analytics in Formula 1. Case context: 2024 Italian Grand Prix; driver context: VER.

Core data

- FastF1
- LapTimeSeconds
- Compound
- TyreLife
- Position

Architecture

Area	Details
Ingestion	Apache Kafka
Processing	Apache Flink / stream processing
Architecture	Kappa
Models	XGBoost, Bi-LSTM, Temporal Fusion Transformer (TFT)
Explainability	TimeSHAP
Imbalance handling	SMOTE

Architecture idea

- Kafka acts as a streaming ingestion/buffering layer.
- Flink/stream processing forms the processing core.
- Kappa treats the stream as the central processing path.
- Data can be partitioned by car identity to preserve temporal order.
- Sliding windows can convert telemetry into feature vectors.

Predictive models

XGBoost

Tree-based predictive model included for high-frequency/state or anomaly-oriented processing.

Bi-LSTM

Sequence model included for temporal relationships in race data.

Temporal Fusion Transformer

Advanced temporal architecture included for multi-horizon temporal forecasting.

Additional concepts

- CAP/CAL-style distributed-system reasoning
- Real-time inference and latency constraints
- TimeSHAP
- SMOTE
- Kappa/Kafka/Flink

Team information in earlier material

Ikram Ullah (F2023376208), Mohib Ullah (F2023376225), Section A1; advisor Dr. Rimsha Anam.

3. Human Activity Recognition Using Deep Learning

Deep Learning Final Project — Computer Vision

Area	Details
Status	Documented academic final project
Scope note	The project source explicitly documents UCF101, MobileNetV2 and LSTM. Evaluation/deployment items listed as expected or future work are not presented as guaranteed completed deployment.

Team

Project material lists Ikram Ullah, Mohib Ullah and Lariab Asif at UMT.

Objective

Classify human activities in video by learning spatial information from frames and temporal relationships across frame sequences.

Architecture

Area	Details
Dataset	UCF101
Spatial model	MobileNetV2
Temporal model	LSTM

Framework	PyTorch / Torchvision
Video processing	OpenCV
Evaluation	Accuracy, confusion matrix, precision, recall, F1-score, training curves

Workflow

16. Import libraries
17. Configuration
18. Dataset loading
19. Dataset exploration
20. Preprocessing
21. Frame extraction
22. Dataset preparation
23. DataLoader
24. Baseline model
25. Training
26. Evaluation
27. Fine-tuning
28. Comparison
29. Model saving
30. Future deployment

Libraries

- PyTorch
- Torchvision
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Scikit-learn
- PIL
- tqdm

Applications discussed

- Smart surveillance
- Healthcare monitoring
- Sports analysis
- Human-computer interaction
- Autonomous robots
- Security monitoring
- Elderly fall detection
- Smart homes
- Video analysis
- Fitness tracking

Future deployment direction

The project material lists FastAPI and Docker as future deployment targets.

4. KRR for Robotics — KnowRob, RoboBrain & CRAM

Knowledge Representation and Reasoning for Robotics

Area	Details
Status	Academic presentation / research project
Scope note	The presentation documents the conceptual architecture and reported focus. Detailed experimental methodology and quantitative results were not preserved.

Problem statement

Robots may perceive objects but still need knowledge representation, reasoning and planning to understand situations and perform tasks effectively.

Overall pipeline

Sensors → Perception → Knowledge Base → Reasoning → Planning → Action

Area	Details
KnowRob	Semantic knowledge framework; stores objects, actions and experiences.
RoboBrain	Large-scale knowledge engine for sharing knowledge across robots.
CRAM	Task planning and execution; handles failures and replanning.

System layers

- Perception layer
- Knowledge layer
- Reasoning layer
- Planning layer
- Execution layer

Reported focus

- Planning
- Reasoning
- Object understanding
- Task execution

5. Amazon Reviews NLP Assignment

Natural Language Processing — Kaggle Notebook

Area	Details
Status	Academic NLP coursework
Scope note	Dataset statistics and preprocessing results are documented. This section does not claim a production sentiment-analysis deployment.

Dataset

Area	Details
------	---------

Environment	Kaggle Notebook
Path	/kaggle/input/datasets/ikramwazir/amazon-reviews-dataset/Amazon_Reviews.csv
Text column	Text
Rows	21,214
Unique documents	20,407
Average character length	460.843
Missing Text	159
Unique raw tokens	50,037
Unique cleaned tokens	32,907
Reduction	34.23%

Preprocessing

- Cleaning/normalization
- Tokenization
- Stopword handling
- Preserving not/no/nor/never
- Stemming
- Lemmatization
- TF-IDF / IDF
- Statistical language models / unigram
- Information retrieval
- Text similarity
- Naive Bayes
- Confusion matrix

Deliverables

- Assignment1.ipynb
- cleaned_dataset.csv
- README.md

Recorded issue

The contractions module was missing in the Kaggle environment during the work.

6. Ikrambot — Personal RAG Chatbot

NLP Semester Project / Personal Knowledge Base

Area	Details
Status	Current project — data collection and organization phase
Scope note	The current phase is collecting and structuring personal/custom information. The later RAG implementation is a separate development phase.

Purpose

Ikrambot is the personalized chatbot identity for the NLP semester project. Its current purpose is to collect, organize and document personal information for use as a custom RAG dataset.

Dataset categories

- Personal profile
- Education
- Courses studied
- Technical skills
- Projects
- Business and entrepreneurship
- Goals and future plans
- CV and cover-letter material
- Professional communication portfolio
- Research notes and technical interests

Planned RAG pipeline

31. Load documents
32. Preprocess
33. Chunk
34. Generate embeddings
35. Store in a vector database
36. Retrieve relevant chunks
37. Pass retrieved context to an LLM
38. Prompt engineering
39. Conversation-history maintenance
40. Chatbot interface
41. Online deployment

Data-quality principle

Known facts, historical information, planned goals and exploratory ideas are kept distinct. Missing information is marked rather than invented.

Portfolio value

The project connects NLP, information retrieval, Generative AI and RAG while using a personal/custom dataset.

7. Online Event Management & Ticket Booking System

Software Engineering Semester Project

Area	Details
Status	Academic software project
Scope note	The available record documents the software-engineering deliverables and system focus. Detailed implementation language, database schema and deployment details were not preserved.

Purpose

Manage events and support ticket-booking workflows through a user-oriented software system.

Documented deliverables

- SRS documentation

- Use case diagrams
- Class diagrams
- Sequence diagrams

System focus

- Event workflows
- User interaction
- Ticket booking
- User-friendly interfaces

8. eCommerce Inventory Management System / DSA Projects

C++ Data Structures and Algorithms

Area	Details
Status	Academic lab / mini-project work
Scope note	The records support an inventory-management/DSA context and the listed data structures, but not a complete production eCommerce specification.

Data structures

- Arrays
- Linked lists
- Stacks
- Queues

Learning outcomes

- Applying theoretical DSA concepts in code
- Implementing core operations on linear data structures
- Connecting DSA with an inventory-management context
- Improving algorithmic problem solving

9. Robotics & Embedded Systems Projects

Arduino / ESP32 / Mobile Robot / Sensor Experiments

Area	Details
Status	Exploration, lab work and technical prototypes
Scope note	This section consolidates multiple robotics experiments. It is not presented as one single finalized commercial robot.

Hardware

- Arduino Uno
- ESP32
- ESP32-CAM
- L298N
- L293D / AFMotor shield

- 4-wheel chassis
- 4 DC motors
- 18650 battery holder/cells
- HC-SR04
- IR sensors
- MPU6050
- GPS NEO / NEO-6M
- HC-06
- Servo
- MP3 module and remote

Concepts explored

- Collision avoidance
- Bluetooth control
- Wi-Fi control
- IR/path following
- GPS robotics
- ESP32-CAM/video concepts
- Swarm robotics concepts

Troubleshooting

- L298N power setup without a buck converter
- MPU6050 reported 'not found' despite LED activity
- ESP32 package 2.0.17 used in a context where newer ledcAttach behavior caused compatibility problems

10. RTL-SDR & Weather Satellite Exploration

Software-Defined Radio / Satellite Reception

Area	Details
Status	Technical exploration and experimentation
Scope note	The records document hardware/software exploration and technical interests, not a verified commercial or permanent satellite ground station.

Equipment

Area	Details
SDR	RTL-SDR R820T2 USB tuner
Computer	Windows
Software	SDR#
Satellite software	Satsuma v1.2.2
Antenna	V-dipole

Satellite interests

- Meteor-M
- NOAA-18
- NOAA-19

Topics

- Weather-satellite image reception
- SDR fundamentals
- Radio direction finding
- Signal monitoring
- Lawful GPS signal-health/interference-monitoring concepts

11. Professional Communication Portfolio

EN-220 Communication Skills Final Project

Area	Details
Status	Academic communication project
Scope note	The project was designed around a five-minute professional online presentation and a collection of workplace communication documents.

Portfolio components

- Resume
- Cover Letter
- Professional Email
- Business Letter
- Questionnaire
- Abstract
- Report
- Reflection

Presentation formats

- Live online presentation
- Narrated PowerPoint
- Canva voice-over

Skills

- Professional writing
- Business communication
- Document preparation
- Presentation structure
- Professional self-introduction

12. Hospital LAN Design

Earlier Academic Project

Area	Details
Status	Limited-detail project record
Scope note	Only the project title is preserved in the available project history; detailed topology, addressing, devices, protocols and results were not retrieved.

Known information

- Project title: Hospital LAN Design.
- Domain: computer networking / LAN design.

No additional technical details are asserted because the available project source does not support them.

13. MyFitGoal Flutter Fitness App

Flutter Application

Area	Details
Status	Limited-detail project record
Scope note	The available record identifies the project as a Flutter fitness application but does not preserve its complete feature list, database design or deployment details.

Known information

- Project title: MyFitGoal.
- Technology/context: Flutter / Dart.
- Associated with beginner-level Flutter application work.

No additional technical details are asserted because the available project source does not support them.

14. Human Pose Estimation

Computer Vision / AI

Area	Details
Status	Limited-detail project record
Scope note	Human pose estimation appears in earlier project history, but model, dataset, framework and evaluation details were not preserved.

Known information

- Topic: Human Pose Estimation.
- Domain: Computer Vision / AI.

No additional technical details are asserted because the available project source does not support them.

15. GAN Research

Generative Adversarial Networks

Area	Details
Status	Limited-detail project record
Scope note	GAN research appears in earlier project history, but the exact architecture, dataset, experiments and results were not preserved.

Known information

- Topic: GANs.

- Domain: Deep Learning / Generative AI research.

No additional technical details are asserted because the available project source does not support them.

Cross-Project Technical Skills

Area	Details
Programming	Python, C++, Dart, SQL
Machine Learning	TF-IDF, LinearSVC, LabelEncoder, cosine similarity, Naive Bayes, Decision Trees, Linear Regression, XGBoost
Deep Learning	ANN, RNN, LSTM, Bi-LSTM, MobileNetV2 + LSTM, TFT
NLP	Preprocessing, tokenization, stopword handling, negation handling, stemming, lemmatization, TF-IDF/IDF, statistical language models, information retrieval, text similarity, RAG
Data / analytics	NumPy, Pandas, Matplotlib, FastF1
AI systems	Streamlit, document processing, Gmail integration, RAG concepts
Data engineering	Kafka, Flink, Kappa architecture, TimeSHAP, SMOTE
Embedded	Arduino, ESP32, ESP32-CAM, sensors, GPS, Bluetooth, motor control
Radio / SDR	RTL-SDR, SDR#, Satsuma, V-dipole, satellite-reception concepts
Software engineering	SRS, use-case, class and sequence diagrams
Tools	Jupyter, Google Colab, Kaggle, VS Code, Anaconda, MySQL, Android Studio, FlutLab, Code::Blocks

Project Development Progression

42. Programming and Python/C++ foundations
43. DSA and eCommerce/inventory mini-projects
44. Software Engineering event-management system
45. Flutter/MyFitGoal exploration
46. Machine Learning and AI projects
47. Deep Learning and Computer Vision — Human Activity Recognition
48. Robotics and embedded systems
49. RTL-SDR and weather-satellite exploration
50. Formula 1 data warehouse and predictive analytics
51. AI resume screening
52. NLP and Amazon Reviews coursework
53. Generative AI and RAG learning
54. Ikrambot personal-data collection

Portfolio Data-Quality Notes

- Completed academic projects, research/exploration, current work and limited-detail project names are distinguished.

- Formula 1 advanced real-time architecture contains research/proposed system elements and should not automatically be described as a fully deployed live motorsport platform.
- Robotics and RTL-SDR sections consolidate multiple experiments rather than claiming one finished commercial product.
- Ikrambot is currently in the personal-data collection/organization phase; the full RAG implementation is a later phase.
- Where a source did not provide a metric, implementation detail or result, this document does not invent one.

Source Basis

Compiled from the user's project documents, personal dataset files, CV/project records and academic project materials available in the conversation/library. The Human Activity Recognition and KRR for Robotics projects were included because separate project files were found in the user's available materials.